

Building People with Three-Dimensional Memory

Why and how we need to exercise our memory muscles



Coquelicots (In Flanders Fields) by Robert Vonnoh, 1890

There's a sense now that something [didn't happen if you don't share it](#). There are young people who wouldn't understand going to an event, travelling somewhere, being in a relationship, if they couldn't post about it. They would not see the point. They simply cannot conceive of a life that exists without an audience consuming it.

from [You Don't Have To Document Everything](#) by Freya India

The fear that photographs may steal away our soul may sound like spiritual hyperbole, but given that people around the world take [5.3 billion](#) photos daily, sharing them with friends, family, and countless strangers online, we may indeed wonder whether we are losing part of our self in the process. The incessant distraction of interfacing with devices leaves us feeling as if our brain and our body are forever in a different place. It almost seems as if we are in a race to upload our life into the virtual universe. Our desire to capture and share the present is numbing our ability to form natural memories of the moments we want to actually treasure. By excessively documenting our lives artificially Marshall McLuhan might say we are “autoamputating” our memory.

When we use our devices as memory keepers, we not only interfere with the formation of long-term memories, but we also flatten our experience and personal identity into a one-dimensional digitized

version of ourselves. Long before outrage over the [Apple commercial](#)'s crushing of our treasured cultural objects, we had already begun abdicating our minds to machines. But there is an alternative: We can chose to strengthen what over-reliance has withered and build three-dimensional memories.

Why flex our memory ? - by Peco

The “Shass Pollacks” memorized the entire Babylonian Talmud. The book is thousands of pages, and some of these memory masters knew every word, even its location, with pinpoint accuracy. Literally. According to a [report](#) from 1917,

a pin would be placed on a word, let us say, the fourth word in line eight; the memory sharp would then be asked what word is in this same spot on page thirty-eight or fifty or any other page; the pin would be pressed through the volume until it reached page thirty eight or page fifty or any other page designated; the memory sharp would then mention the word and it was found invariably correct. He had visualized in his brain the whole Talmud; in other words, the pages of the Talmud were photographed on his brain.

There were no USB sticks in 1917, which means that the Shass Pollaks were like human hard drives. Yet their accomplishment was more than a memory feat. They were carrying the collective spiritual mind of the Jewish people through history.

The Shass Pollaks weren't entirely unique. We see similar feats in [Hinduism](#), [Islam](#), and [Christianity](#), and outside of religion, as in the case of one actor who [memorized all 60,000 lines](#) of Milton's Paradise Lost. Impressive as all this might be, it might also seem pointless in an online era when we have access to almost all factual information at our fingertips. Why bother memorizing anything, if I know I can find it on Google or a USB stick? In fact, if I know I can find it, why bother looking for it in the first place?

A growing memory loss is creeping through our society. Not a total memory loss, but a loss of motivation for our participation in the act of memory¹. Even an apathy about it. Where will this lead?

In the earliest stages of Alzheimer's—a disease of memory loss—people are sometimes aware that something isn't right. Familiar words fail to come to mind. Recent conversations are harder to retrieve. People become understandably concerned. Then—if it really turns out to be the actual disease—what follows next is a more worrisome stage in which memory progressively worsens, even to the point that

people begin to forget they have a problem. They forget that they are forgetting. At this stage, any concern or sadness of losing one's memory dissipates. It is not possible to be sad about losing your memory, if you no longer realize that you are losing your memory.

Is there some analogy to a society losing its collective memory through cognitive offloading? We might expect some initial concern about the loss, yet also a growing indifference as the concern fades into the background. Some segments of the culture might experience a carefree insouciance as they become largely forgetful of what has come before—the wisdom, knowledge, and traditions of history—and more gripped by the here-and-now stimulation of their screens.

As real Alzheimer's progresses, there is not only memory loss, but disorientation, anxiety, depression, aggressive behavior, and delusional thinking. Perhaps when the memory loss of a society becomes great enough, we see something similar. We might find ourselves wondering, *Why do so many people suddenly have such strange ideas? Why are they so afraid? Or so angry?*

When we start to lose our grip on the larger whole of life, we can only cling to the fragments. Usually the emotionally powerful bits. And we will tend to cling to these bits with pathological fixity, as deep down we know that we aren't on secure ground—or else we cling so fixedly because the delusion is so great that we believe we *are* on secure ground. It's a more optimistic delusion, yet still a delusion.

It might be stretching the analogy too far to suggest that Western society is in the grip of a digital Alzheimer's disease; but it would be stretching too little to say that there is no analogy at all, and to glibly suppose, as some do, that we can knock away the pillars of our shared human responsibility for remembering without any major consequence. And it would be wildly naive to believe that our machines can take over this role. People with severe memory problems often live in nursing homes surrounded by a support system of attendants. Nursing homes are a blessing to those who need them, but not to those who don't. Reliance on a support system of machines to hold our collective memories is a formula for docility. When Steve Jobs brought us Apple computer [we were promised](#) bicycles for the mind, but many of us feel we're ending up with cognitive wheelchairs.

And now Meta/Facebook wants schools to use its virtual reality headset to help children learn². Can technology fix a memory problem that technology is helping to cause? The Silicon Valley optimists will cheerily insist it can. But the real concern has little to do with education. The real concern is the principle

of [ubiquitism](#)—one of Big Tech’s unspoken operating rules—that new technology must be available for pretty much anything, anywhere, anytime, for anyone. And so we create better and better cognitive wheelchairs for our kids. Can we blame them if they feel so comfortable they don’t want to get out of the chairs, or don’t trust the ground under their own feet?

Freya India’s [observation](#) that “there’s a sense now that something didn’t happen if you don’t share it” hints at an existential crisis, a feeling that we aren’t real if we aren’t posting online. A feeling that the memories which might have stabilized our minds, our identities, have turned to quicksand, and now we’re floundering, now we’re trying to solve our insecurity by gripping more tightly to the very technology that’s feeding into the problem.

Once the headsets become ubiquitous, we won’t only feel that we ourselves lack reality if we’re not online. We’ll feel that the real world itself lacks reality. Or that it’s not as real, not as vibrant and interesting, as the virtual.

We still have a choice when it comes to technology. We don’t have to accept the ubiquity of the cognitive wheelchairs. We can return to our own minds, and learn and remember the knowledge, wisdom, and skills that are lifegiving. We can be responsible for carrying our little piece of history forward.

Once our brain cells die, they generally don’t regrow, yet the one part of the brain that seems to defy this rule is our [hippocampus](#), which is involved in memory formation and learning. This ought to give us hope, and remind us that we’ve got muscles in our minds.

How to Build Three-Dimensional Memories - by Ruth

If we are to flex our memory muscles then we must strive for deep encoding, engaging multi-faceted modalities in the process. We are three-dimensional beings with five senses that the digital medium does not even come close to approximating. Our motivated memory links to things we care about: love is one of the most potent conduits of memory. Thus, the more we pour our heart and undivided attention into the moments we want to treasure, the deeper the memories will grow into our marrow.

What follows are practical suggestions for deepening personal and educational memory that we have used within our own family.

Photos



My great-grandmother (on the right) with her older sister.

I never met my great-grandmother and there is only a single childhood photograph of her, yet I formed an internal image of her through the many I stories I was told (my favorite being her hand-feeding a sick chicken on her lap in her farmhouse kitchen). Today, children's existence is recorded as soon as they are born, some parents going as far as posting their [newborn's first 48 hours](#) onto social media. This excessive documenting not only cheapens the memory, but also teaches children from the very start that the world is to be viewed and valued through a screen.

Many of you will remember the days of the 24-exposure film roll, or the seemingly extravagant 36 version. There were limits. It was necessary to consider what pictures were worthwhile taking, and the captured moments were rendered more precious. The average person now carries nearly [3000 photos](#) on their phone which can send us off into dizzying scrolls in cyberspace, whereas photos that are printed and collected into tangible albums can serve as storytelling tools in a shared present.

- **Decide that less is more.**
- **Choose to simply enjoy some moments and events without taking photos and instead be fully present in them.**
- **If you take pictures, treasure them rather than sharing them with strangers online.**
- **Commit yourself to a small selection of photos for a physical album.**

Journals



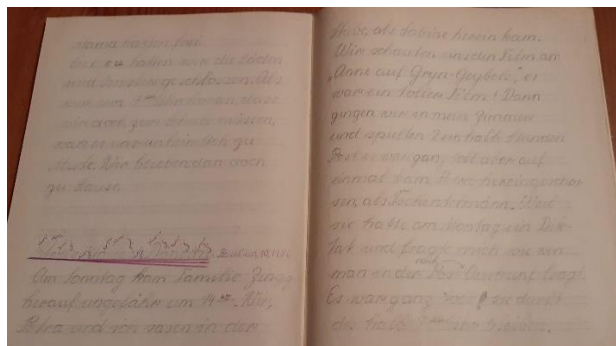
Here are some of the journals around our house. The blue one with a hand-made cover was my first diary, used weekly at school.

"Preserve your memories, keep them well, what you forget you can never retell."

- Louisa May Alcott

Journaling at the end of the day or week helps to translate events into words. The act of writing experiences down by hand engraves them further in our memory and has the added benefit of processing out emotions. Before we had children I kept quite detailed diaries, collecting cinema stubs, receipts, or other paper mementos. As a young mother entries became shorter and less frequent, but one practice that I found especially useful was writing down children's quotes that made me laugh (e.g. "Do roly polies have family relations?"; "What do you want to be when you grow up? - A rocket!").

Paging through old diaries can reawaken memories that are seemingly lost, and more than an image, recapture the mood and mind of an earlier age.



In these pages my ten-year old self recounts watching "Anne of Green Gables" for the first time, an inspiration that would lead me to move to Canada in my later teen years.

Building People with Three-Dimensional Memory by Peco and Ruth Gaskovski 6

Our oldest daughter reminded me this week that when she was sixteen, she wrote a diary entry every single day for one whole year. Some of the entries simply listed what she did during the day or who she saw, others had some deeper reflections. When I asked her whether she experienced any effect from this practice, she noted that the days felt more distinct and time felt less like a blur. When we fail to pay attention to the details of our days, they not only bleed into each other monotonously, they are also less memorable.

- **Get a physical journal that you enjoy holding in your hand.**
- **Keep your journal visible and easily accessible.**
- **Instead of scrolling, decide to pick up a pen and simply start by noting down some of the events of the day, even if they seem banal.**
- **Keep writing, even if you can't do it regularly.**

Storytelling



The Picknick by Emile Claus, 1887

A beautiful, sacred memory, preserved from childhood, is perhaps the best education. If a man stores up many such memories to take into life, then he is saved for his whole life. And even if only one good memory remains with us in our hearts, that alone may serve someday for our salvation.

Alyosha, from Dostoyevsky's Brothers Karamazov

Storytelling is as old as humanity itself. While few of us get to sit around a fire telling tales, we all still build our lives through narratives that shape our view of the world and form our memories. You don't have to be an expert raconteur to reminisce, share favorite memories, and retell family stories.

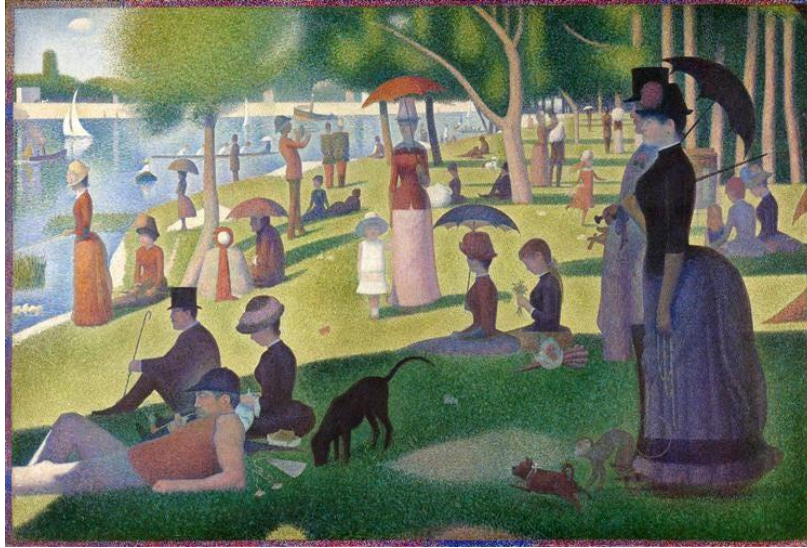
It was actually a spontaneous storytelling session with our children around the kitchen table one evening that prompted us to want to write this piece on growing deeper memories, ones that are seemingly baked into our bones.

It started with me reading some funny quotes from one of my diaries which made us laugh tears, which led to recounting details around the events, which in turn prompted more and more shared remembrances until an hour had passed and it was time to head to bed.

Interestingly the storytelling not only engrained these memories more deeply, but it left us feeling engaged with each other, flushed with good feeling, and resulted in a greater willingness to be helpful. It felt like the memories served to glue us together.

- **Spend some time after dinner, on a walk, or even while driving recalling some favorite memories with your children, spouse, or parent. If you live alone, call up a family member or friend simply to reminisce.**
- **If you have an elderly parent ask them to retell some of their favorite childhood memories.**
- **If you have children, spend time recalling the events of the day, the week, and the month. Have them retell their favorite happenings, asking younger ones questions to help them fill out details.**
- **Flipping through photo albums together is a great storytelling starter.**

Training Visual Memory



Sunday Afternoon on the Island of La Grande Jatte by Georges Seurat, 1884

The true art of memory is the art of attention.

- Samuel Johnson

Sitting next to the porcelain-tiled stove in my grandmother's Swiss kitchen, I used to lay out a large array of memory game cards and challenge her to a round. She was 80, I was 8, so I would invariably win with a huge stack, proud of my apparently superior memory. To this day I remember that the card with a small, scratched dot on the back was an reddish-yellow apple and the one with a little black mark in the corner was an oak leaf.

We are inundated with a barrage of visual images wherever we turn, but we rarely pay deliberate attention to the details we are seeing. For children, playing memory card games helps them to slow down and focus on visual details.

My mother, who we see once a year on our visits to Switzerland, took the time to create a personalized photo memory game for her grandchildren. This set not only includes favorite snapshots of the holiday, but also photos of all the cousins, aunts, and uncles. Creating such personalized sets is an excellent method to help your children deepen family memories.



Picture study is another wonderful method to train visual memory. If you live near a gallery, take advantage of experiencing original paintings in all their detail. We visit the “Kunstmuseum” in my hometown Basel whenever we are there and love revisiting the brush strokes of Monet, van Gogh, Pissarro, and Holbein.

At home you can display an art print, examining it daily. By the end of the week turn it over and try to recall as much detail as possible. Seurat actually painted a study of “La Grande Jatte” devoid of any people (see below). You may have thought that you are very familiar with his famous painting (see beginning of section), but how many details can you actually recall?



La Grande Jatte by Georges Seurat, 1884

Learning and Memory

Our ability to find humor in the world, to make connections between previously unconnected notions, to create new ideas, to share in a common culture: All these essentially human acts depend on memory.

Joshua Foer, USA Memory Champion

Beyond the practices that deepen personal memory, we can steer away from our reliance on machines by developing habits that allow our biological memory to find a firm foothold. This requires patience, diligence, and time. But anything worthwhile does.

We may be under the impression that some people simply have great memories and other don't, just like Joshua Foer once thought. Foer set out as a freelance journalist to witness competitors memorize strings of numbers, card sequences, and poems at the USA memory championships. This experience so captivated him that he went on to train himself in an ancient memory technique, entered as a competitor the following year, and promptly won the championship³.

Anyone can train their memory, especially children.

Historically students memorized times tables, poems, the presidents, capitals, Latin conjugations, and famous speeches. This practice is not an outdated mechanical activity as some educators would have you believe, but instead forms the foundation for critical and analytical thinking. The age old trivium model of education, which focuses on memorization especially during elementary years, is making a comeback with [the resurgence of classical education](#)⁴.

Here are some practical examples of memory practices that I have completed with our own children and the students I have taught, including facts, famous speeches, and poetry.

Facts



*A collector's cabinet filled with paintings, shells, coins, fossils and flowers (detail)
by Frans Francken the Younger, 1619*

For memorizing facts, there are few strategies more enjoyable, fun, and stunningly effective than the link-and-story method. Memorize Academy has perfected this approach by joining it with whiteboard animation. This technique uses “visualization and association to leverage the astonishing natural power of visual memory”. One of the fourth grade students in my homeschool chemistry co-op told me, “I know the first 30 elements of the periodic table like I know my name!”

I have tried this with students as young as grade 3 all the way to high school and it worked perfectly for all of them. Here is a sample video of the [‘How to Memorize the Periodic Table’](#) to give you a taste for the link-and-story method. With this method you can memorize the entire periodic table in around three hours. You can access many of the videos for [free on the site’s blog](#), including presidents, longest rivers, pi, and more. Once you get the idea of the method, you can easily develop your own version of the link-and-story method for facts you would like to memorize. [The Craft of Memory](#) also has excellent resources on [how to memorize the Bible](#), [remembering names](#), and [regions, capitals, and cities](#).

Speeches

Memorizing famous speeches has been a practice since the ‘Golden Age of Eloquence’ in ancient Greece. The following is a collection of excerpts of great speeches I compiled for students to memorize including, Theodore Roosevelt, Abraham Lincoln, Susan B. Anthony, Elie Wiesel, Martin Luther King, Mother Teresa, Indira Ghandi, Elizabeth I, and more. Here are the [Excerpts of Famous Speeches for Students to Memorize Download](#).

Poetry



In his article [Why We Should Memorize](#) which appeared in *The New Yorker*, Brad Leithauser presents a profound argument for how poetry changes not just our brain, but also our heart:

The best argument for verse memorization may be that it provides us with knowledge of a qualitatively and physiologically different variety: you take the poem inside you, into your brain chemistry if not your blood, and you know it at a deeper, bodily level than if you simply read it off a screen.

One of the poems my homeschool co-op students memorized was *A Psalm of Life* by Henry Wadsworth Longfellow. We used the [Mensa for Kids Poem #10 template](#). I then broke the poem down into [three practice lessons](#), which left students to fill in the missing words. After several rounds of practice and dedication, the students were able to recite the complete poem.

You can find more tips on how to memorize poetry and a list of 10 classic poems to commit to heart on [Mensa's A Year of Living Poetically](#), including ["Invictus"](#) by William Ernest Henley, ["Sonnet 116"](#) by William Shakespeare, and ["Do Not Go Gentle into that Good Night"](#) by Dylan Thomas.

Our closing reflection

At the end of the Apple commercial, the crushing machine rises, and in place of the myriad objects that it's flattened, we see a sleek thin piece of glass that is the new iPad. But it might as well be a metaphor for the flattening of our minds.

If we exchange our three-dimensional memories—memories formed through motivated and multisensory engagement with the real world—for screen-based memories, we won't just end up with withered memory abilities, but we'll become thinner human beings who feel less substantial and less secure in themselves, and whose experience of being "real" will become increasingly dependent on devices.

One of the things we like to tell our kids is to "use your brain to bits". We have been given the most extraordinary ability to experience and remember reality, free of charge. Why would we choose to forgo this gift?



Plums Blossom by Claude Monet, 1879

What is worth remembering for you?

Do you have any memory strategies?

Do you practice any memory work yourself or with your kids?

We would love to hear from you! Please share your thoughts, reflections, and questions in the comments section!

Further Reading

[Recommended Books and Resources on the Art of Memory](#) by Ronald Johnson

[Why I'm Memorizing the Divine Comedy by Dante Alighieri](#) by Ronald Johnson

[Recovering Our Memory](#) by Michael Toscano in First Things

[You Don't Need To Document Everything](#) by Freya India. When a post receives well over 3000 likes and 727 restacks, you know that it has hit a nerve.

[There Is No Thinking without Memorizing](#) by Jon D. Schaff in Public Discourse

Did you hear that we are going on [A Pilgrimage out of the Machine](#)? We would love to have you join us! We will be leading an eleven-day pilgrimage on the Camino trail in Spain next year, from June 14-24, 2025. Joining us, as co-leader, will be writer/photographer Seth Haines from [The Examine](#). Please Note: Space is limited so reserve your spot promptly :)

